CHEMISTRY | YEAR 10

BONDING – METALLIC BONDING

KEY VOCABULARY

Metallic bond: the strong attraction between positive metal ions and delocalised electrons.

Delocalised: electrons that are free to move through the whole structure.

Lattice: a regular, repeating arrangement of ions.

Malleable: can be hammered or bent into shape without breaking.

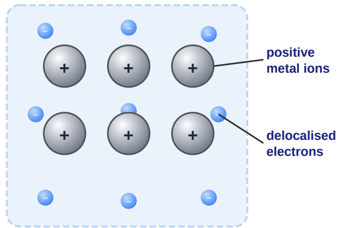
COMMON MISCONCEPTIONS

~~✗ Metals are made of molecules.~~ → **✓ Metals are a giant lattice of ions.**

~~✗ The electrons belong to one atom.~~ → **✓ Electrons are delocalised (shared by all).**

~~✗ Metals conduct because ions move.~~ → **✓ The delocalised electrons carry the charge.**

THE METALLIC LATTICE



Positive **metal ions** sit in a 'sea' of **delocalised electrons**.

1 DEFINE THE BOND

Describe what is meant by **metallic bonding**.

Use the words 'positive metal ions' and 'delocalised electrons'.

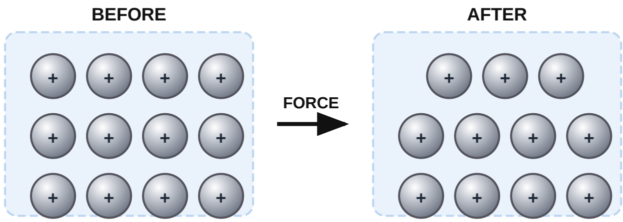
2 ELECTRICAL CONDUCTIVITY

Explain why metals are good conductors of electricity.

Think about which particles are free to move.

3 MALLEABILITY

The diagram shows the lattice before and after a force is applied. **Explain** why the layers of ions can slide.



4 MELTING POINT

Complete the sentence using the word bank.

Metals have ______ melting points because there is a ______ electrostatic ______ between the ions and the delocalised electrons.

high	strong	attraction

TIP: *Positive metal ions are held in a 'sea' of delocalised electrons.*

✓ Think: Why does a bigger ionic charge give a stronger metallic bond?